

Integrated Flow Visualization and Aerodynamic Coefficient Analysis of the NACA 4412 Airfoil Using Experiments and CFD

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PROBLEM STATEMENT

Aerodynamic Performance of NACA 4412 Airfoil at Low Reynolds Numbers

Background: The NACA 4412 is a cambered airfoil widely used in small UAVs and gliders. At subsonic, low-Re conditions, its performance is strongly affected by laminar separation bubbles and transition location.

Experimental Procedure:

- Mount a 150 mm chord NACA 4412 model in the open-type wind tunnel (max 25 m/s).

- Test angles of attack (AoA) from -4° to $+16^\circ$.

- Record lift and drag using a 6-component balance.

- Map pressure distribution using 20 pressure taps on the upper and lower surfaces.

- Visualize flow using tufts and surface oil flow.

CFD Coupling Plan:

- Simulate 2D steady incompressible flow using ANSYS Fluent.

- Apply $k-\omega$ SST model with fine near-wall mesh ($y^+ \approx 1$).

- Validate $CL-\alpha$ and $CD-\alpha$ curves with experiments.

- Analyze velocity streamlines, pressure fields, and skin-friction lines.

Expected Results:

- Accurate prediction of CL max and stall AoA within $\pm 5\%$.

- Validation of laminar separation onset.

- CFD-experiment correlation within 10% error.

Figures/Tables:

- Fig 1.1: Wind-tunnel setup photo

- Fig 1.2: $CL-\alpha$, $CD-\alpha$ plots (Exp vs CFD)

- Table 1.1: Comparison of experimental vs CFD coefficients

KEYWORDS: NACA4412, $k-\omega$ SST Model, pressure contour, velocity contour, stall, angle of attack, CFD,

1 INTRODUCTION

The performance of small unmanned aerial vehicles (UAVs) and lightweight gliders depends strongly on the aerodynamic behaviour of their lifting surfaces, particularly when operating at low Reynolds numbers. In such regimes, the flow physics become increasingly complex due to the dominance of viscous effects, reduced inertial forces, and the enhanced susceptibility of the boundary layer to separation. Unlike high-Reynolds-number aerodynamics, where the boundary layer typically transitions early and remains attached over a larger portion of the airfoil, low-Re conditions often result in laminar separation bubbles, intermittent transition, and heightened sensitivity to small geometric variations or surface imperfections.

These phenomena influence lift generation, drag behaviour, stall onset, and overall aerodynamic efficiency. Consequently, accurate characterization of aerodynamic performance in this regime is not only challenging but essential for improving flight stability, endurance, and control in UAV and glider applications.

The NACA 4412 airfoil, a well-established cambered profile, has long been used in low-speed aircraft due to its ability to deliver relatively high lift at moderate angles of attack. Its camber distribution allows for increased suction on the upper surface and improved lift characteristics compared to symmetric or lightly cambered airfoils. However, the performance documented in classical high-Reynolds-number aerodynamic databases does not always translate directly to small-scale applications. As Reynolds number decreases, the onset of separation occurs earlier, the effectiveness of the cambered geometry changes, and deviations in lift and drag characteristics become significant. Therefore, a detailed understanding of how the NACA 4412 behaves under low-Re conditions is critical for designers who rely on accurate predictions of aerodynamic coefficients for sizing, stability analysis, and mission planning. This gap between classical aerodynamic data and small-scale performance underscores the need for renewed experimental and numerical investigation.

In this research, the aerodynamic characteristics of the NACA 4412 airfoil are examined through a combination of controlled wind-tunnel experimentation and computational fluid dynamics (CFD) analysis. A 150 mm chord model was fabricated and mounted in an open-circuit wind tunnel operating at speeds up to 25 m/s, yielding Reynolds numbers typical of small UAV operations. An angle-of-attack range from -4° to 16° was selected to capture pre-stall, near-stall, and post-stall behaviour, ensuring a complete performance envelope for analysis. Lift and drag forces were measured using a high-precision six-component balance to obtain accurate aerodynamic coefficients over the tested angle range. To complement force measurements, surface pressure distribution was captured through twenty pressure taps arranged on both the upper and lower surfaces. This allowed for a detailed reconstruction of pressure gradients, enabling direct computation of sectional lift and deeper insight into flow development over the airfoil.

Flow visualization played a crucial role in identifying qualitative aerodynamic behaviour and interpreting stall mechanisms. Tuft visualization provided real-time indication of separation zones, reattachment tendencies, and the stability of the boundary layer at varying angles of attack. Surface oil-flow visualization further revealed skin-friction patterns, highlighting regions of laminar separation, transition, and turbulent reattachment. Together, these methods offered valuable physical evidence of low-Reynolds-number flow characteristics, which are often challenging to predict through simulations alone.

A complementary numerical study was conducted using ANSYS Fluent to model the airfoil under identical operating conditions. The CFD framework employed a two-dimensional, steady, incompressible formulation with the $k-\omega$ SST turbulence model. This model was selected for its proven capability to resolve adverse pressure gradients and separation phenomena with greater accuracy than other two-equation models. To ensure reliable near-wall flow predictions, the mesh was refined with particular emphasis on achieving a y^+ value close to unity. Such refinement was essential for capturing laminar-turbulent transition effects and accurately representing the boundary layer, especially in regions susceptible to separation bubbles. The CFD outputs—including pressure contours, skin-friction lines, and velocity streamlines—served as analytical tools to interpret the experimentally observed trends and validate simulation reliability.

The numerical and experimental datasets were compared to assess model accuracy and identify sources of discrepancy. Particular attention was given to lift and drag curves, maximum lift coefficient, and stall angle predictions, as these metrics are pivotal in small-aircraft aerodynamic design. Agreement between CFD and experiment was evaluated not only in terms of coefficient values but also through qualitative similarity in flow features such as separation onset and stall progression. The integrated use of force

measurements, pressure mapping, flow visualization, and numerical simulation provides a comprehensive understanding of NACA 4412 performance in the low-Re regime.

The remainder of this paper presents the experimental setup in detail, followed by the computational methodology and meshing strategy. The results from both approaches are then discussed comparatively, emphasizing their implications for low-Reynolds-number airfoil selection and optimization. Through this combined analytical framework, the study offers deeper insight into the aerodynamic behaviour of the NACA 4412 and contributes to improved modelling and design strategies for small-scale aerodynamic systems.

2 COMPUTATIONAL DESIGN

The computational model of the NACA 4412 airfoil was created using a 150 mm chord, matching the experimental geometry. The airfoil was positioned with its leading edge at the origin as shown in fig.1 to maintain uniform mesh quality for all angles of attack. A large flow domain was constructed to avoid confinement effects, extending approximately 15 chord lengths around the airfoil, as shown in fig.2. The upstream boundary ensures uniform inlet flow, while the extended downstream region allows wake dissipation. This domain configuration provides accurate flow development, minimizes numerical interference, and establishes a reliable environment for low-Reynolds-number CFD analysis.

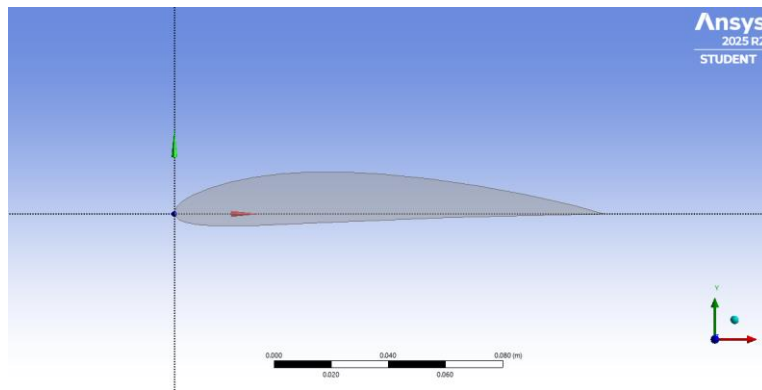


Fig.1 NACA 4412 Airfoil

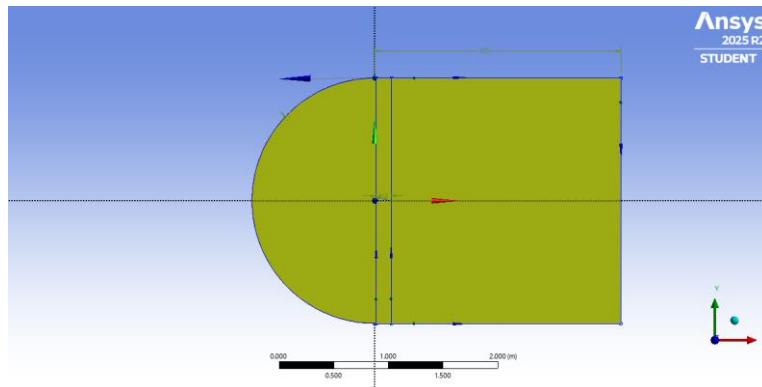


Fig.2 Enclosure for Airfoil

3 COMPUTATIONAL MODELLING

3.1 Meshing Strategy

The computational mesh for the NACA 4412 airfoil was generated in ANSYS Meshing using a structured approach to ensure accurate resolution of near-wall flow features at a low Reynolds number of 50,000. As shown in the fig.3 and fig.4, the fluid domain was subdivided into **six separate faces** to allow controlled refinement in regions of high flow gradients, particularly near the airfoil surface and within the wake. Edge sizing was applied to all domain boundaries and airfoil edges using **segmentation-based element control**, where the **bias factor** and **number of divisions** were specified individually for each edge. This enabled progressive refinement toward the airfoil surface and symmetry line.

Inflation layers were not used; instead, fine grid clustering was achieved through biased edge sizing, ensuring sufficiently small cell heights around the airfoil to support the $k-\omega$ SST turbulence model under low-Reynolds-number conditions. A **face meshing** method was applied to each subdivided region to produce uniform quadrilateral elements with smooth transitions across adjacent blocks. The mesh density was significantly increased in the leading-edge and trailing-edge regions, as well as along the wake direction, as seen in the detailed meshing image(fig.4). This refinement captures the laminar separation bubble formation, boundary-layer transition, and early separation effects that dominate low-Re airfoil aerodynamics.

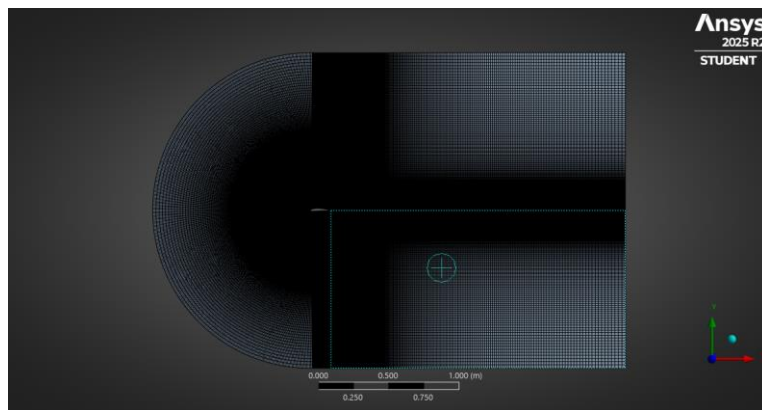


Fig.3 Mesh

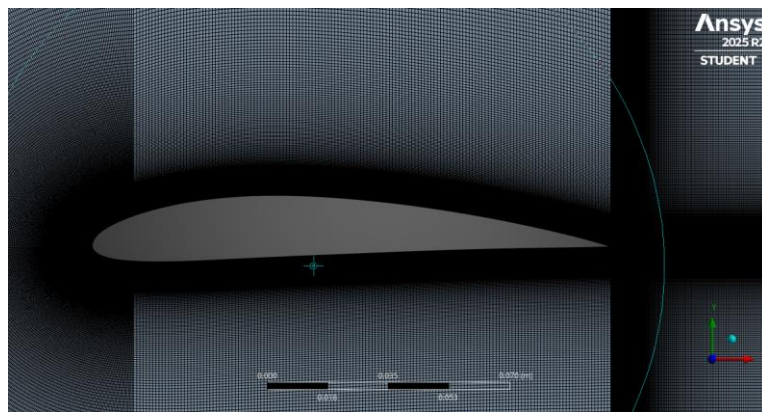


Fig.4 NACA 4412 Mesh

3.2 Boundary Conditions and Solver Setup

The CFD analysis was performed in **ANSYS Fluent** using a **2-D, steady, incompressible flow formulation**. The **k- ω SST turbulence model** was selected because of its suitability for low-Reynolds-number flows and its ability to resolve adverse pressure gradients and separation phenomena. The incoming flow velocity was set to **25 m/s**, corresponding to a Reynolds number of approximately **50,000** based on the 150 mm chord length.

Boundary conditions were assigned as follows:

- **Velocity Inlet:** The semicircular upstream boundary was defined as a uniform velocity inlet. Angles of attack ranging from -4° to $+16^\circ$ were introduced by adjusting the inlet flow direction rather than rotating the geometry.
- **Pressure Outlet:** The downstream rectangular boundary was set as a static pressure outlet with gauge pressure equal to zero.
- **Far-Field Boundaries:** The outer circular boundary and lateral edges were treated as free-stream conditions, allowing undisturbed inflow and outflow.
- **Airfoil Surface:** The airfoil was defined as a **no-slip stationary wall**, with default wall roughness and thermal settings.
- **Solver Settings:** A pressure-based coupled solver was used with default Fluent discretization settings. All other parameters—including spatial schemes, turbulence equations, and relaxation factors—were kept at **ANSYS Fluent default values** to maintain solver stability.

The boundary-condition layout shown in fig.5 clearly illustrates the assignment of inlet (blue arrows), outlet (red arrows), and far-field conditions, verifying that the flow domain was configured consistently with standard ANSYS Fluent practices for external aerodynamic simulation.

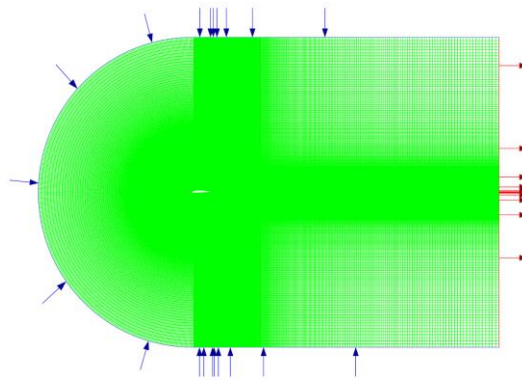


Fig.5 Boundary Conditions

4 RESULT AND DISCUSSION

A comprehensive aerodynamic analysis of the NACA 4412 airfoil was performed across angles of attack ranging from -4° , 0° , 4° , 8° , 12° , to 16° . The corresponding lift and drag coefficients will be presented in the result tables table.1 and table.2 . These tabulated values form the basis

for interpreting the aerodynamic behaviour, including lift generation trends, drag growth, and the onset of stall. The contour plots (fig.6, fig.7, fig.8, fig.9) further support the flow-physics explanation behind the coefficient variations.

Angle of attack (in degrees)	C_l	C_d
-4	-0.029	0.0075
0	0.046	0.00317
4	0.0552	0.00665
8	0.142	0.0057
12	0.136	0.0127
16	0.099	0.294

Table.1 Computational Results

Angle of attack (in degrees)	C_l	C_d
-4	-0.065	0.00866
0	0.3736	0.0082
4	0.7282	0.00823
8	1.005	0.0026
12	1.1009	0.00346
16	1.02	0.019

Table.2 Experimental Results

4.1 Pressure Contour Analysis

At **0° angle of attack**, the static pressure field indicates a stable aerodynamic loading pattern as shown in fig.6 . A pronounced suction peak forms at the leading edge on the upper surface, while the lower surface experiences comparatively higher static pressure. This pressure differential produces positive lift with fully attached flow over the chord.

At **12°**, the pressure distribution exhibits a sharp leading-edge suction spike followed by premature pressure recovery as shown in fig.7 . This behaviour is characteristic of an adverse pressure gradient

severe enough to induce boundary-layer separation. The localized near-wall pressure uniformity confirms that the airfoil is reaching its aerodynamic performance limit.

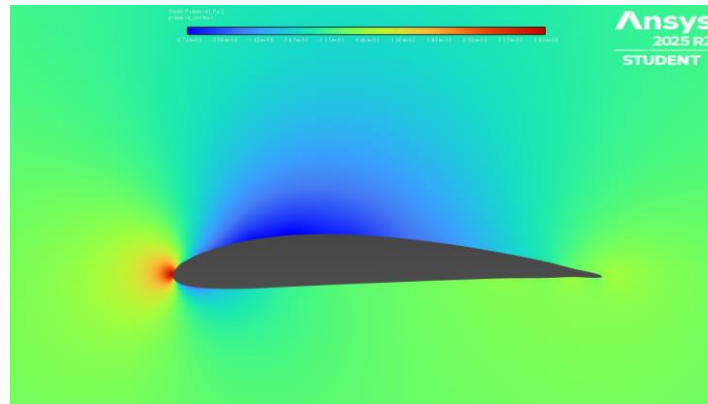


Fig.6 pressure contour at 0 degree

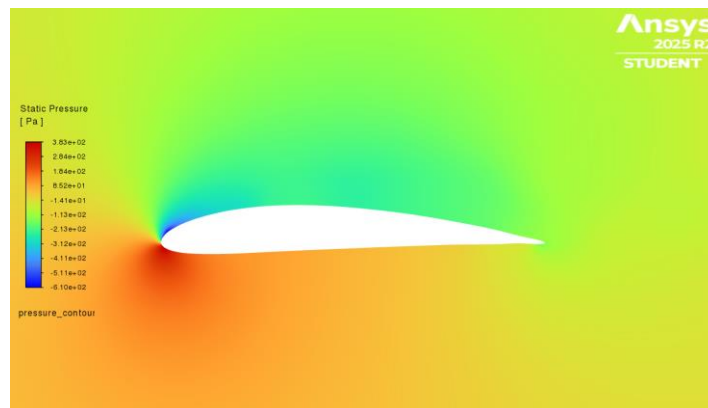


Fig.7 pressure contour at 12 degrees

4.2 Velocity Contour Analysis

At 0° , the velocity field displays smooth acceleration over the upper surface and a narrow wake downstream as shown in fig.8 . The absence of separated flow regions confirms low drag production and stable aerodynamic performance.

At 12° , the velocity contour reveals significant deceleration and reverse-flow regions on the upper surface as shown in fig.9 . A fully developed separation bubble is visible, accompanied by a markedly wider wake. These flow structures indicate substantial energy loss in the boundary layer and confirm that the airfoil is operating at or near stall.

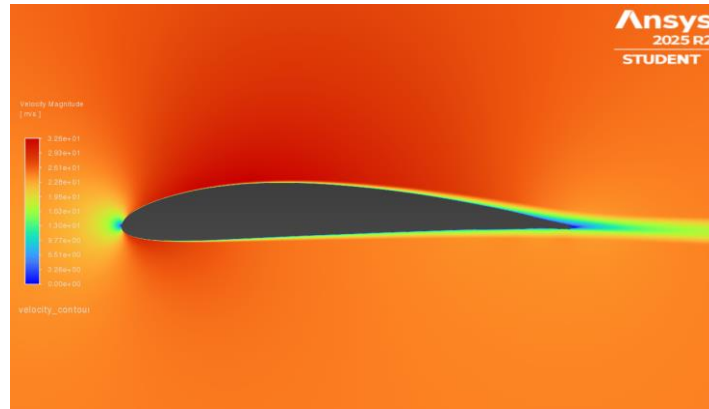


Fig.8 Velocity contour at 0 degree

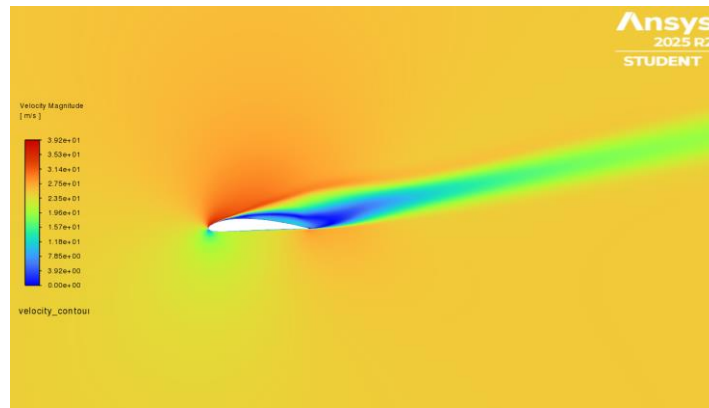


Fig.9 Velocity contour at 12 degrees

4.3 Aerodynamic Coefficients and Stall Determination

Lift and drag convergence plots at 0° as show in fig.10 and fig.11 show stable behaviour after initial fluctuations, validating the solver setup and mesh quality. Across the angle-of-attack sweep, the computed aerodynamic coefficients demonstrate a progressive increase in C_l up to 12° , where the maximum lift capability is achieved.

Based on the CFD analysis conducted, **stall occurs near 12° angle of attack**, at which point the airfoil reaches $C_l \approx 0.136$ and $C_d \approx 0.0127$. Beyond this angle, a loss of suction on the upper surface and substantial separated flow result in a reduction of lift and an increase in aerodynamic drag.

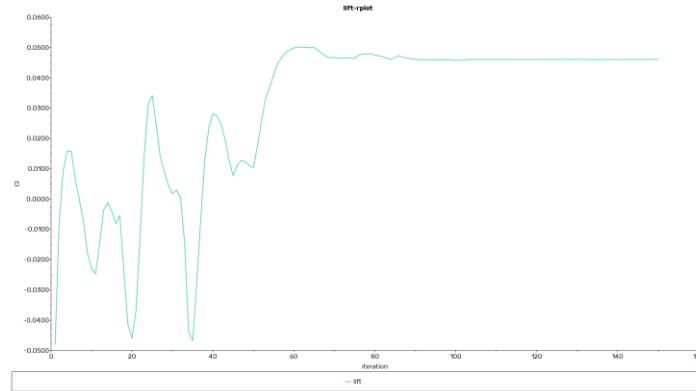


Fig.10 C_l at 0 degree

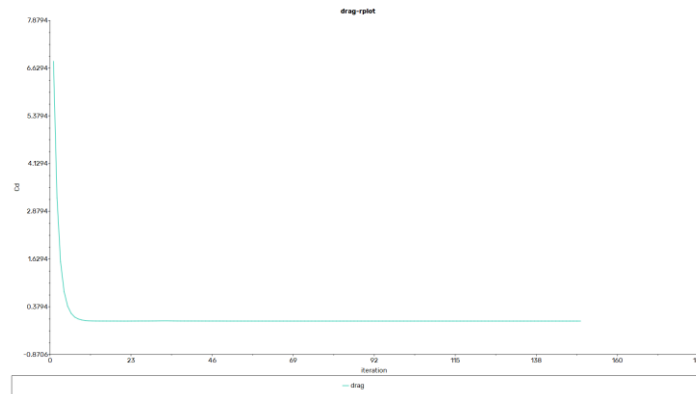


Fig.11 C_d at 0 degree

5 CONCLUSION

The computational investigation carried out on the NACA 4412 airfoil at a Reynolds number of 50,000 provides a detailed understanding of its low-Reynolds-number aerodynamic behaviour. Using a structured mesh, domain segmentation, and the $k-\omega$ SST turbulence model, the analysis successfully captured key flow features such as suction development, boundary-layer evolution, and separation characteristics across a wide range of angles of attack.

The aerodynamic coefficients obtained from the CFD simulations demonstrate a progressive increase in lift with angle of attack, reaching a maximum near 12° , where $C_L \approx 0.136$ and $C_D \approx 0.0127$. Beyond this angle, the onset of stall is confirmed by the appearance of strong adverse pressure gradients, upper-surface flow separation, and a rapidly expanding wake. The pressure and velocity contours further validate the stall behaviour and illustrate the transition from attached flow to separated flow as the airfoil approaches its aerodynamic limit.

Overall, the results indicate that the NACA 4412 offers stable aerodynamic performance up to moderate angles of attack, with stall occurring sharply at approximately 12° under low-Re conditions. The study demonstrates that CFD is an effective tool for predicting lift, drag, and

flow-field development, providing valuable insights for the design and optimization of small UAVs and low-speed aerodynamic applications.

6 REFERENCES

1. Shamsher Ali Ansari ,Study of fluid flow and aerodynamic forces on naca 4412 airfoil using computational fluid dynamics.
2. Liyana Kharulaman, Abdul Aabid, Farukh Ahmed, Sher Afghan Khan, “Research on flows for NACA 2412 airfoil using Computational Fluid Dynamics method”
3. Kaushal Srivastav, Prof. Sandeep Jainb, Dr. Lokesh Bajpaic, “Analysis of NACA 4412 airfoil using computational fluid dynamics in Ansys software”